



DATTA MEGHE
INSTITUTE OF HIGHER
EDUCATION & RESEARCH
(DEEMED TO BE UNIVERSITY)
LEARN LEAD

FACULTY OF
ENGINEERING AND
TECHNOLOGY
(A CONSTITUENT UNIT OF DMIHER DU)

4th Cycle
Re-Accrediation
NAAC A++
(CGPA 3.78)



nirf
RANKING 2025

49th
University
in India

20th
Medical

36th
Dental

84th
Overall

11-50
SDG
Institution
Band

“3rd DMIHER(DU) International Conference on Artificial Intelligence in Healthcare, Education and Industry” (IDICAIHEI-2025)

NOVEMBER 28-29, 2025

Conference Id - 65991





DATTA MEGHE
INSTITUTE OF HIGHER
EDUCATION & RESEARCH
(DEEMED TO BE UNIVERSITY)
LEARN LEAD

**FACULTY OF
ENGINEERING AND
TECHNOLOGY**
(A CONSTITUENT UNIT OF DMIHER DU)

4th Cycle
Re-Accrediation
NAAC A++
(CGPA 3.78)



nirf
RANKING 2025

49th
University
in India

20th
Medical

36th
Dental

84th
Overall

11-50
SDG
Institution
Band



Track: AI In Healthcare

Paper ID: 276

Paper Title: Advancing Non-Invasive Scoliosis Detection Using Deep Learning

Author Name: Devanshu Ekhar

**Co-author(s) Name: Jeena Joseph, Sidhant Lohkar, Saveri Dongre,
Dr. Yogesh Golhar**

IDICAIHEI-2025
Conference Id - 65991
(Hybrid Mode Conference)





Outline of the Presentation

- 1. INTRODUCTION**
- 2. METHODOLOGY**
- 3. RESULTS AND DISCUSSIONS**
- 4. CONCLUSION**
- 5. REFERENCES**





SECTION 1: INTRODUCTION



Background

- **Scoliosis:** A 3D spinal deformity causing lateral curvature and vertebral rotation.
- Affects **2–3% of children aged 10–16 years.**
- If undiagnosed, can lead to chronic pain, lung restriction, and posture imbalance.
- Early detection = vital for timely intervention and reduced surgical risk.



Comparative X-ray showing a normal spine vs. a scoliosis-affected spine.



Problem Statement

- Manual scoliosis diagnosis relies on **radiologists and Cobb angle measurement**.
- Limited availability of experts in rural areas → **delayed or missed detection**.
- X-ray interpretation is **subjective and time-consuming**.
- There's a need for an **automated, accessible, and affordable** screening system.





Motivation

- AI and Deep Learning (CNNs) can identify curvature patterns automatically.
- Enables **faster and more consistent** scoliosis screening.
- **Bridges healthcare gaps** in under-resourced communities.
- Potential for integration into **school health programs and telemedicine**.





Research Gap

➤ Existing studies use CNNs for detection, but:

✗ Lack **real-time, deployable systems**.

✗ Require expensive hardware or depth sensors.

✗ Not optimized for low-resource setups.

→ **Our Solution:** A lightweight, accurate **CNN + Streamlit Web Application** for real-time scoliosis detection.





Objectives

1. Develop a CNN model to detect scoliosis from spinal X-rays.
2. Train and test it on a **publicly available dataset**.
3. Build a **Streamlit web interface** for real-time use.
4. Validate model accuracy and usability.
5. Promote **non-invasive, accessible screening tools** for healthcare.





Literature Review (1996 – 2011)

Author (Year)	Focus	Key Finding	Limitation
Evans et al. (1996)	MRI in juvenile scoliosis	Early spinal structure studies	No AI approach
Ramirez et al. (1997)	Back pain in scoliosis	Prevalence link established	Clinical, not predictive
Justice et al. (2003)	Genetic susceptibility	Identified X-linked gene factor	Small sample size
Mo & Cunningham (2011)	Pediatric scoliosis review	Summarized treatments	No automation proposed





Literature Review (2019 – 2021)

Author (Year)	Approach/Model	Key Findings	Gap
Yang et al. (2019)	CNN on back images	86% accuracy	Used back photos, not X-rays
Kokabu et al. (2021)	3D CNN with depth sensor	High accuracy	Costly equipment
Wang et al. (2021)	DL on X-rays	84% accuracy	No web deployment
Fraiwan et al. (2021)	Dataset creation	259 labeled X-rays	Small dataset
Sung et al. (2021)	Statistical study	Nationwide scoliosis trends	No ML use





Literature Review (2022 – 2024)

Author (Year)	Approach/Model	Key Findings	Gap
Fraiwan et al. (2022)	Transfer learning	80.5% accuracy	No deployment
Nguyen et al. (2022)	CNN-LSTM	Multi-view curve detection	Computationally heavy
Yadav et al. (2023)	CNN on ortho X-rays	Good generalization	Not scoliosis-specific
Zhang et al. (2023)	Segmentation CNN	High precision	No telehealth link





SECTION 2 : METHODOLOGY





Proposed Work Overview

➤ Goal:

To design a **CNN-based scoliosis detection system** trained on X-ray images and deploy it through a **Streamlit web app** for real-time usability.

➤ Key Components:

1. Data preprocessing
2. CNN architecture
3. Model training and evaluation
4. Streamlit integration





Dataset Description

- Source: Mendeley Data (Fraiwani et al., 2021).
- Total images: 259 spinal X-rays
 - i. 188 scoliosis cases
 - ii. 71 normal spines
- Publicly labeled dataset ideal for training CNNs.
- **Challenges:** Small dataset & class imbalance.





Data Preprocessing Steps

1. Convert to grayscale
2. Resize to **224×224 pixels**
3. Normalize pixel values $\rightarrow [0, 1]$
4. Encode labels: 0 (Normal), 1 (Scoliosis)
5. Split into **training (80%)** and **validation (20%)**



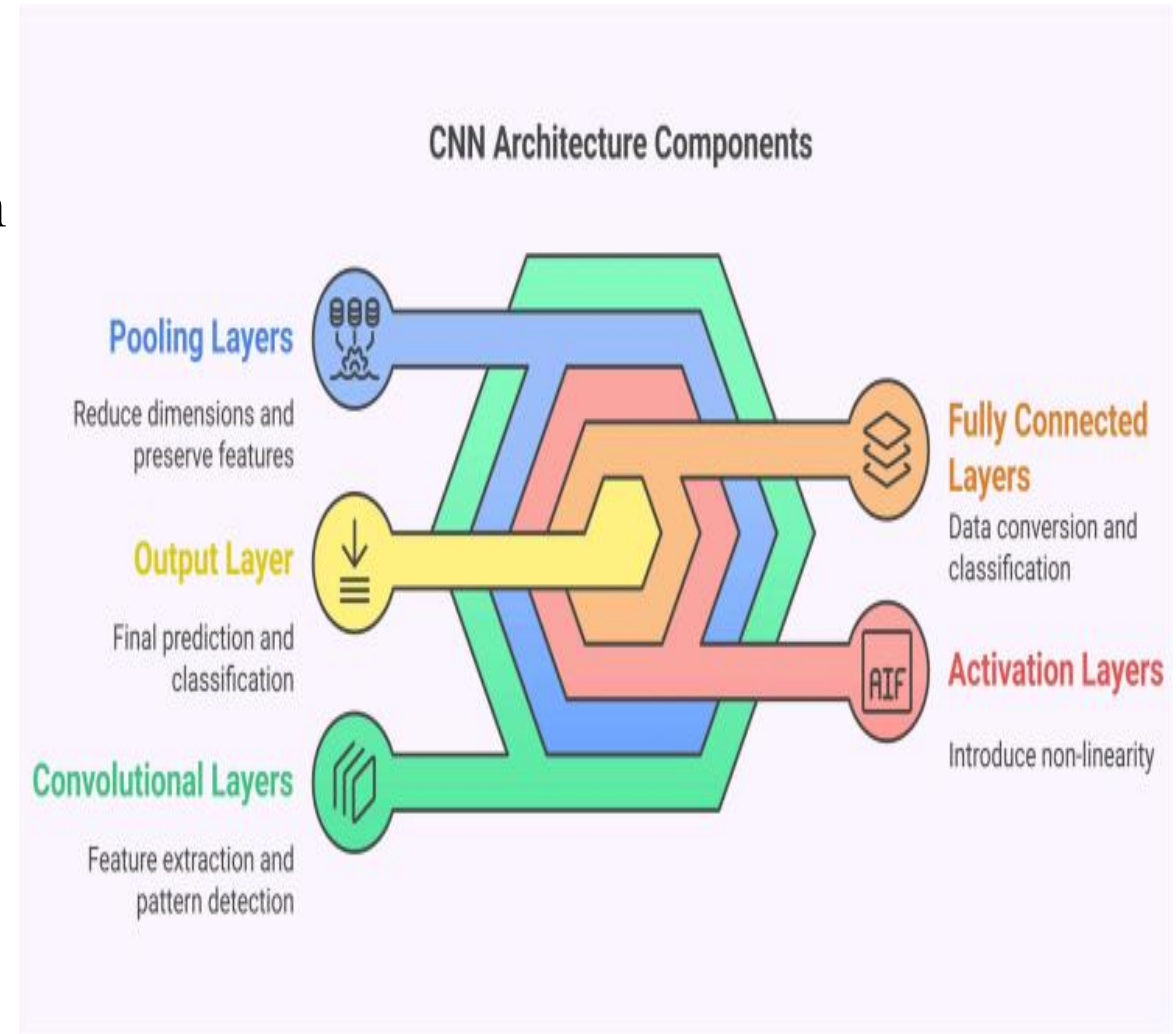
CNN Model Architecture

➤ Model Layers:

- Convolutional → feature extraction
- MaxPooling → dimensionality reduction
- Dropout → reduce overfitting
- Dense Layers → classification
- Sigmoid → binary output

➤ Hyperparameters:

- Optimizer: Adam | LR: 0.001
- Loss: Binary Cross-Entropy
- Epochs: 10 | Batch: 16





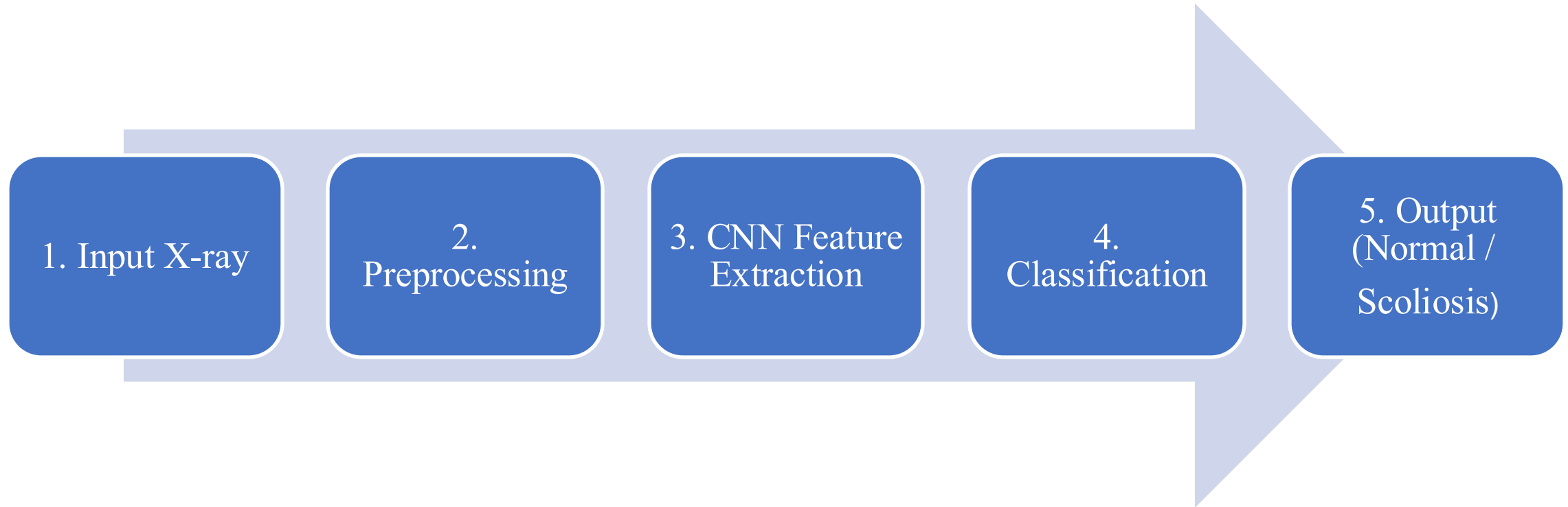
Training Environment

- Platform: **Google Colab (GPU)**
- Libraries: TensorFlow, Keras
- IDE: PyCharm for integration
- Deployment: Streamlit



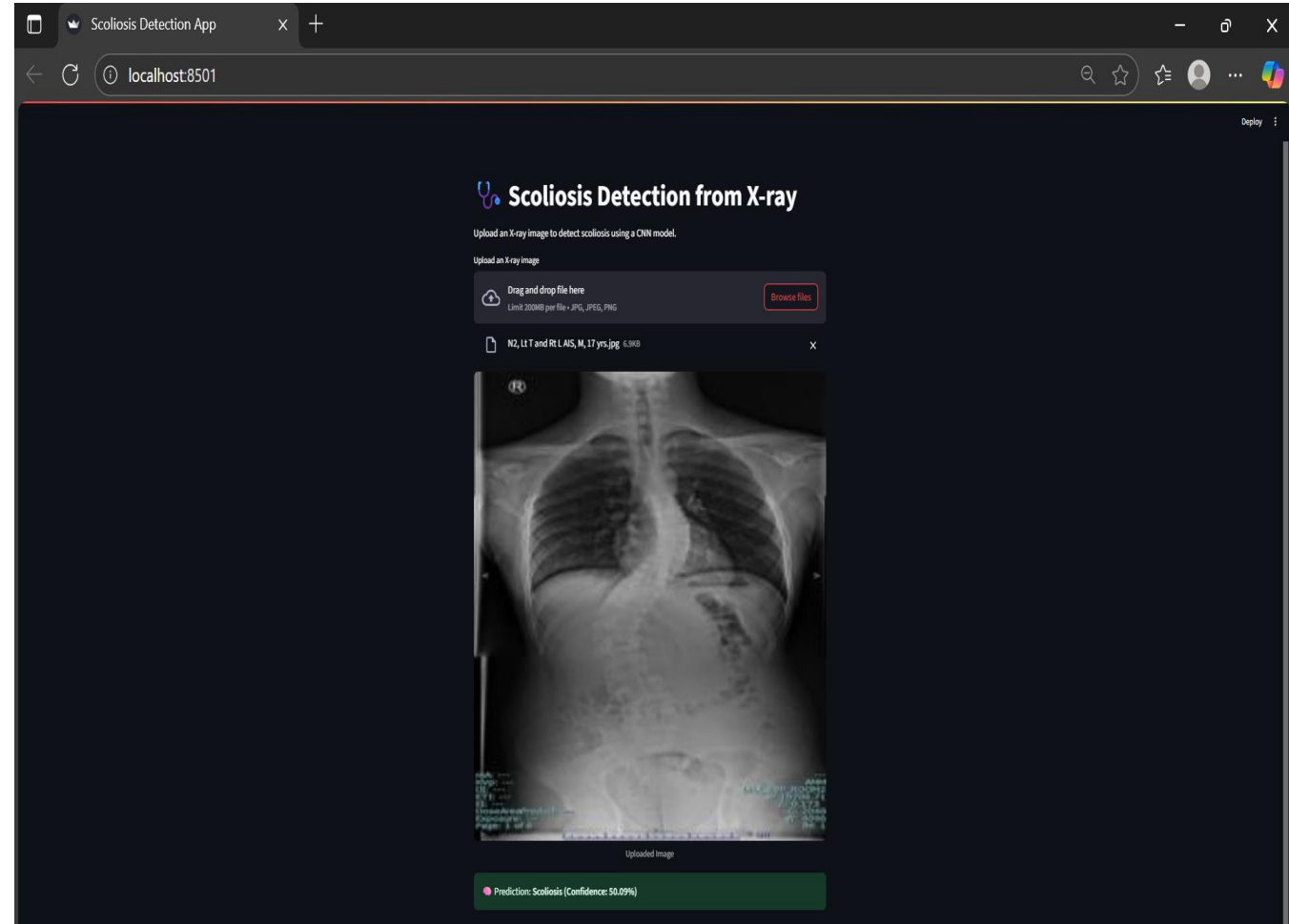
Workflow Diagram

➤ System Pipeline:



Web Application (Prototype)

- Developed using **Streamlit** for user interaction.
- Upload X-ray → get immediate prediction.
- Real-time processing; lightweight interface.
- Deployed locally & accessible for rural screening.





Prototype Demonstration

➤ **Click Here :**

<https://drive.google.com/file/d/1gVI7Ag3UTu4f4lnA8FBD2KwFLDaDVgMZ/view?usp=sharing>

➤ **Narration points:**

- User uploads spinal X-ray
- Model predicts scoliosis instantly
- Clean, accessible, and fast interface
- Ideal for use in schools, telehealth, and small clinics





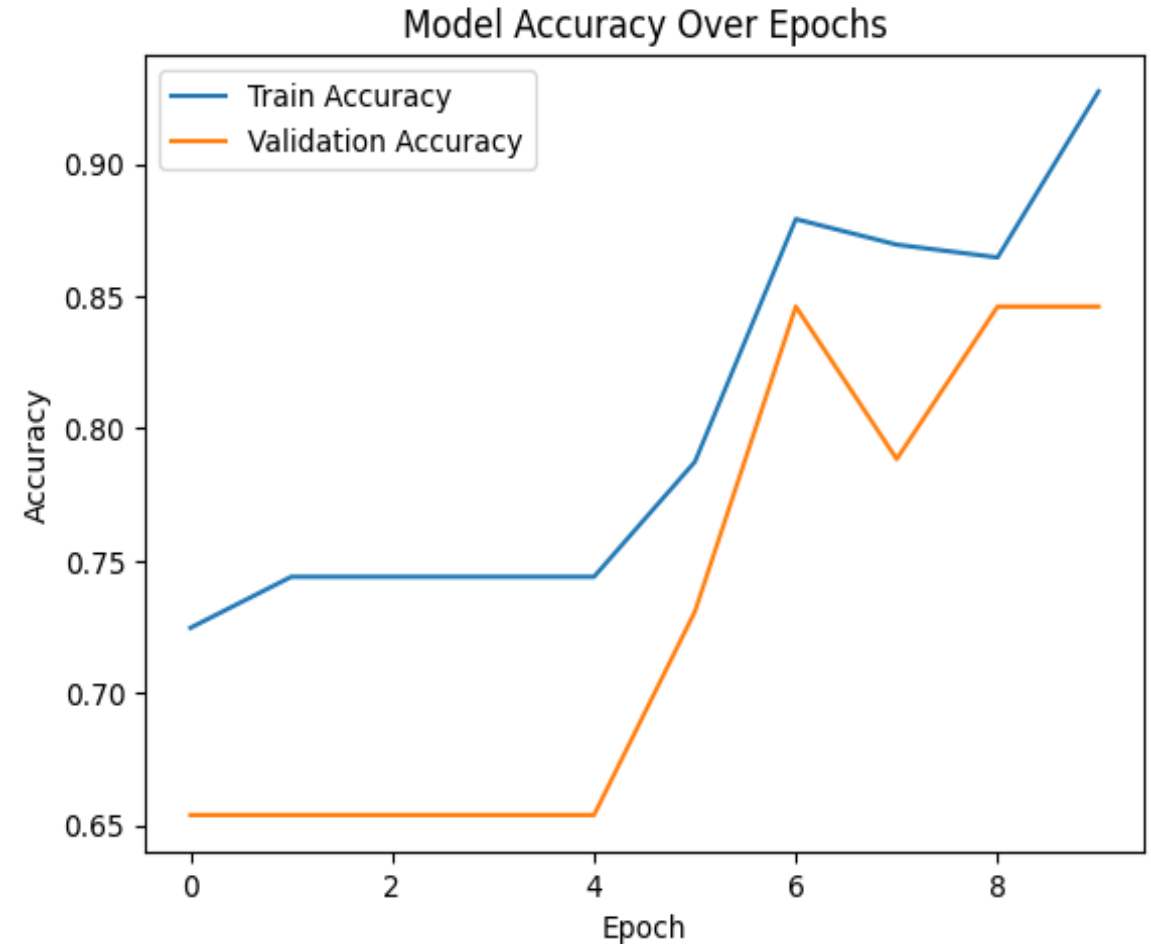
SECTION 3 : RESULTS & DISCUSSION



Model Performance

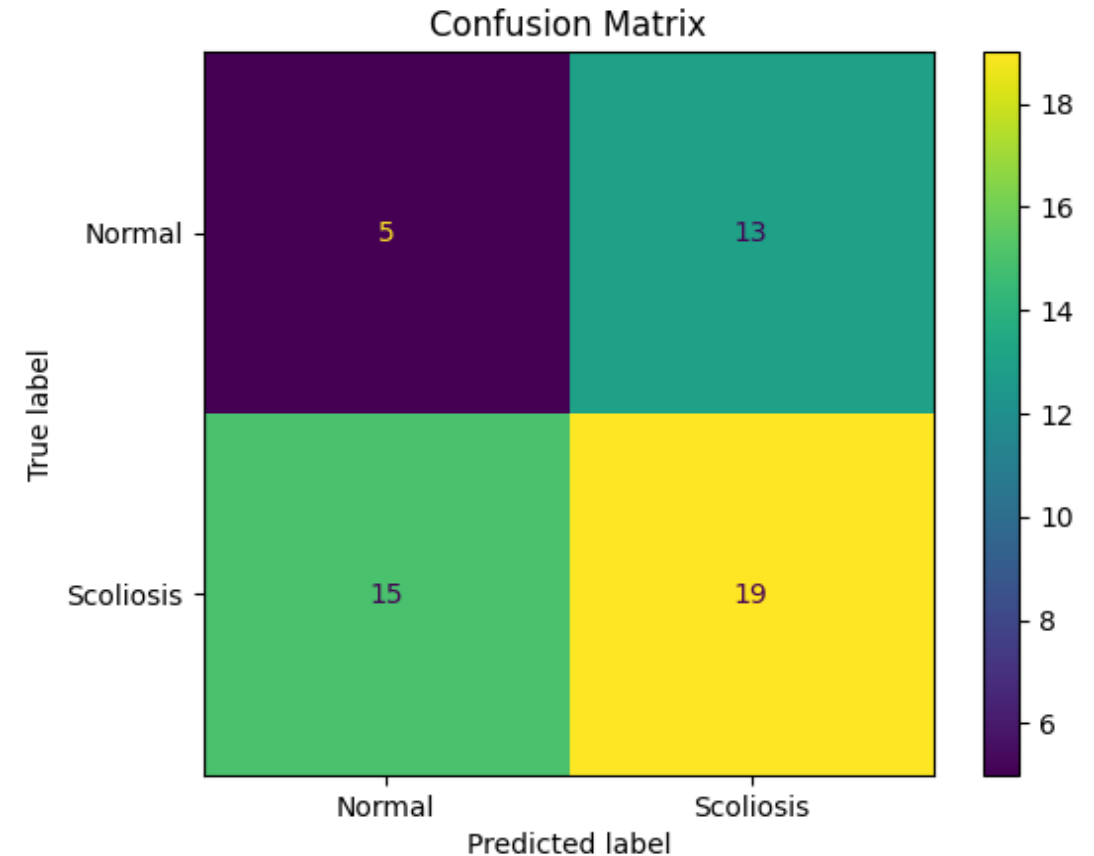
Metric	Value
Training Accuracy	88.65%
Validation Accuracy	78.85%
Validation Loss	0.4723

→ Indicates strong learning despite limited data.



Confusion Matrix Analysis

- **TP:** Correct scoliosis predictions
 - **TN:** Correct normal predictions
 - **FP/FN:** Few errors due to imbalance
- Model performs better on scoliosis class.



Confusion matrix illustrating the classification performance of the scoliosis detection CNN model.



Comparison with Previous Studies

Study	Accuracy	Key Feature
Yang et al. (2019)	86%	Back images
Fraiwan et al. (2022)	80.5%	Transfer Learning
Kokabu et al. (2021)	3D Sensor	Expensive setup
Proposed (2025)	78.85%	Web-enabled, low-cost, real-time system





Key Observations

- CNN effectively learned curvature patterns.
- Limited data → slight accuracy trade-off.
- Web deployment bridges **research–application gap**.
- Acts as **radiologist assistant**, not replacement.





Limitations

1. Dataset imbalance (188 vs. 71 images).
2. Binary classification only.
3. No Cobb angle or severity grading.
4. Not clinically validated yet.
5. Dependent on X-ray quality.





SECTION 4 : CONCLUSIONS





Conclusion

- Developed a **non-invasive AI-based scoliosis detection system**.
- Achieved 78.85% accuracy on validation data.
- Integrated into a real-time **Streamlit web application**.
- Supports faster preliminary diagnosis and remote screening.
- Lays foundation for **AI-driven healthcare accessibility**.





Future Scope

- Collect larger, balanced datasets.
- Add **multi-class severity detection (mild/moderate/severe)**.
- Integrate with **3D MRI/CT** data.
- Use **Grad-CAM** for explainable predictions.
- Deploy on **mobile/edge** devices for field use.





Societal Impact

- Enables early detection in **schools and villages**.
- Supports **telemedicine** programs.
- Reduces workload for radiologists.
- Promotes **AI for social good**.





Acknowledgment

- Guide: **Dr. Yogesh G. Golhar**
- Department of Computer Science & Engineering, **St. Vincent Pallotti College of Engineering & Technology, Nagpur**
- Dataset: Mendeley Data (2021)
- Gratitude to **IDICAIHEI–2025 organizers.**





SECTION 5 : REFERENCES





1st References Slide

- [1] S. C. Evans et al., “MRI of ‘Idiopathic’ Juvenile Scoliosis: A Prospective Study,” J. Bone Joint Surg., vol. 78, no. 5, pp. 733–739, 1996.
- [2] N. Ramírez, C. E. Johnston II, and R. H. Browne, “The Prevalence of Back Pain in Children Who Have Idiopathic Scoliosis,” J. Bone Joint Surg. Am., vol. 79, no. 3, pp. 364–368, 1997.
- [3] C. M. Justice, N. H. Miller, B. Marosy, J. Zhang, and A. F. Wilson, “Familial Idiopathic Scoliosis: Evidence of an X-Linked Susceptibility Locus,” Spine, vol. 28, no. 6, pp. 589–594, 2003.
- [4] F. Mo and M. E. Cunningham, “Pediatric Scoliosis,” Curr. Rev. Musculoskelet. Med., vol. 4, no. 4, pp. 175–182, 2011.
- [5] J. Yang et al., “Development and Validation of Deep-Learning Algorithms for Scoliosis Screening Using Back Images,” Commun. Biol., vol. 2, art. no. 390, 2019.
- [6] J. Xie et al., “Deep Convolutional Neural Networks for Automated Spine Curvature Estimation,” Sci. Rep., 2019.
- [7] P. Ali et al., “A Review of Artificial-Intelligence Applications in Spinal Radiography,” Biomed. Eng. Online, 2020.
- [8] M. Arora et al., “Machine-Learning Approaches for Orthopedic Image Analysis,” J. Orthop., 2020.
- [9] T. Kokabu et al., “Deep-Learning CNNs with 3-D Depth-Sensor Imaging for Scoliosis Detection,” Spine J., vol. 21, no. 6, pp. 980–987, 2021.
- [10] H. Wang et al., “Deep-Learning on Spinal Radiographs to Predict Progression in Adolescent Idiopathic Scoliosis,” EClinicalMedicine, vol. 42, art. no. 101220, 2021.
- [11] S. Sung et al., “Incidence and Surgery Rate of Idiopathic Scoliosis: A Nationwide Database Study,” Int. J. Environ. Res. Public Health, vol. 18, no. 15, 2021.
- [12] M. Fraiwan, Z. Audat, and T. Manasreh, “A Dataset for Scoliosis Detection from X-Ray Images,” Mendeley Data, dataset, 2021.





2nd References Slide

- [13] S. Hasan et al., “AI Methods for Spinal Disorder Detection: A Comparative Study,” Heliyon, 2021.
- [14] A. Albahli, “Deep Learning for Medical Image Classification: Applications and Challenges,” Diagnostics, 2021.
- [15] R. Das et al., “AI-Assisted Diagnostic Tools in Orthopedics—A Review,” J. Healthc. Eng., 2021.
- [16] M. Fraiwan et al., “Using Deep Transfer Learning to Detect Scoliosis and Spondylolisthesis from X-Ray Images,” PLoS ONE, vol. 17, no. 5, 2022.
- [17] L. Pereira et al., “Machine-Learning Methods Applied to Medical Imaging for Spine Disorders,” BMC Med. Imaging, 2022.
- [18] T. D. Nguyen et al., “Hybrid Deep Models for Multi-View Spine Curvature Classification,” Comput. Med. Imaging Graph., 2022.
- [19] Z. Li et al., “Deep-Feature Learning for Spinal Deformity Assessment,” Front. Neurosci., 2022.
- [20] M. K. Gupta et al., “Applied Machine Learning for Scoliosis Detection,” SN Appl. Sci., 2022.
- [21] L. Wang et al., “Sensor- and Image-Based AI Approaches for Spinal Assessment,” Sensors, 2022.
- [22] T. Liu et al., “Computational Approaches for Spinal Curvature Analysis,” Comput. Biol. Med., 2022.
- [23] A. Qureshi et al., “Clinical Validation of AI in Orthopedic Screening,” Cureus, 2022.
- [24] G. B. Lee, D. T. Priefer, and R. Priefer, “Scoliosis: Causes and Treatments,” Adolescents, vol. 2, no. 2, pp. 220–234, 2022.





3rd References Slide

- [25] R. Yadav et al., “CNN-Based Orthopedic Anomaly Detection with Data Augmentation,” IEEE Access, 2023.
- [26] Y. Zhang et al., “Deep Learning for X-Ray Segmentation and Spine Analysis,” IEEE Trans. Med. Imaging, 2023.
- [27] D. Singh et al., “Efficient Scoliosis Detection with Compact CNN Models,” Sci. Afr., 2023.
- [28] P. Kumar et al., “Biocybernetic Methods for Spinal Deformity Classification,” Biocybern. Biomed. Eng., 2023.
- [28] P. Kumar et al., “Biocybernetic Methods for Spinal Deformity Classification,” Biocybern. Biomed. Eng., 2023.
- [29] B. Zhao et al., “Applied-Intelligence Techniques for Spine Disorder Screening,” Appl. Intell., 2023.
- [30] P. K. Das et al., “Recent Advances in Automated Scoliosis Assessment,” J. Healthc. Eng., 2023.
- [31] L. M. Lee et al., “Overview of Current AI Approaches for Spinal Abnormality Detection,” Adolescents, 2023.
- [32] Various Authors, “Recent Trends in AI-Based Scoliosis Detection and Deployment,” Survey/Review, 2023.



Thank You!

Ranked 8th amongst

Health Science Universities in India under “Overall” category

